AI Image Generator

2. Requirements

Version 1.0
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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Stable Diffusion XL + Real-ESRGAN, served locally with Gradio, with per-image GPU telemetry recorded in SQLite.

2. Requirements

2.1 Functional requirements

ID	Requirement	Status
FR-01	Generate an image from a text prompt using a locally hosted diffusion model on the GPU.	Done
FR-02	Provide a browser interface where a prompt can be typed and the result is displayed.	Done
FR-03	Keep the model loaded between requests so repeat generations take seconds, not minutes.	Done
FR-04	Control aspect ratio, resolution, orientation, quality, fidelity, style, count and seed from words in the prompt, with no separate controls.	Done
FR-05	Show how the prompt was interpreted before generation starts.	Done
FR-06	Produce genuine 4K (3840 px) and 8K (7680 px) output when requested, and exact sizes such as 3840x2160.	Done
FR-07	Display, for every image, the GPU used, time taken, VRAM, temperature, power, clocks and software versions.	Done
FR-08	Persist every generation with its settings and telemetry in a database that can be browsed later.	Done
FR-09	Provide a History view with search, aggregate statistics and click-to-view of past images.	Done
FR-10	Save every generated image, native and upscaled, as a PNG file on disk.	Done
FR-11	Support a negative prompt describing what to avoid.	Done
FR-12	Provide command-line equivalents for generation, upscaling and database queries.	Done

2.2 Non-functional requirements

ID	Requirement	How it is met
NFR-01	Runs fully offline after first model download.	Model weights cached in the Hugging Face cache; upscaler weights stored in models/.
NFR-02	A 1 megapixel image generates in under 15 seconds on an RTX 3090.	Measured 4.7 to 13.5 s depending on steps and size.
NFR-03	Peak VRAM stays under 12 GB so 12 GB and 16 GB cards can run it.	Measured peak 10.5 GB including the upscaler.
NFR-04	Upscaling must never exceed VRAM regardless of output size.	512 px tiles with 32 px overlap; memory is bounded per tile.
NFR-05	The server binds to localhost only.	Gradio launched on 127.0.0.1:7860.
NFR-06	All persistent state is in plain files that can be backed up by copying.	PNG files in outputs/, SQLite file generations.db.
NFR-07	The database schema can evolve without losing data.	db.connect() adds missing columns automatically.
NFR-08	Code is dependency-light and readable: no framework beyond Gradio and diffusers.	Six Python modules, about 1,100 lines in total.

2.3 Hardware requirements

Component	Minimum	Recommended	Verified on
GPU	NVIDIA, 12 GB VRAM, compute capability 7.0+	16 GB or more	RTX 3090, 24 GB
System RAM	16 GB	32 GB	-
Disk	15 GB free (7 GB model, 64 MB upscaler, Python environment, outputs)	50 GB for a large output library	294 GB free
CPU	Any x86-64	-	-

Cards below 12 GB can run the command-line script, which enables CPU offload automatically, but the web app assumes the full pipeline fits in VRAM.

2.4 Software requirements

Package	Version used	Purpose
Python	3.12.0	Runtime
torch	2.14.1+cu126	Tensor library with CUDA 12.6
torchvision	0.29.1+cu126	Required by transformers image processors and spandrel
diffusers	0.41.0	Stable Diffusion XL pipeline
transformers	5.20.2	CLIP text encoders used by SDXL
accelerate	1.15.0	Device placement helpers
safetensors	0.8.0	Safe weight loading
gradio	6.30.0	Web interface
gradio_client	2.7.2	Used in tests to call the app programmatically
spandrel	0.4.2	Loads the Real-ESRGAN architecture and weights
pandas	3.0.6	History table in the UI
Pillow	12.3.0	Image I/O, resizing, cropping
numpy	2.5.4	Tile arrays for the upscaler
NVIDIA driver	610.88	Also provides nvidia-smi for telemetry

2.5 Model assets

Asset	Source	Size	Location
stabilityai/stable-diffusion-xl-base-1.0 (fp16 safetensors)	Hugging Face Hub	6.7 GB	%USERPROFILE%\.cache\huggingface\hub
RealESRGAN_x4plus.pth	github.com/xinntao/Real-ESRGAN releases v0.1.0	64 MB	models/ inside the project

2.6 Constraints and assumptions

- Stable Diffusion XL is trained at about 1 megapixel. Native generation is therefore capped at a 1536 px long side; larger sizes are reached by upscaling.
- The telemetry module depends on nvidia-smi being on the PATH, which the NVIDIA driver installer guarantees on Windows. If it is missing, telemetry fields are stored as NULL and generation still works.
- The application is single-user and single-GPU. Concurrent requests are queued by Gradio and processed one at a time.
- The SDXL licence (CreativeML Open RAIL++-M) and the Real-ESRGAN licence (BSD-3) permit local and commercial use with the usual use restrictions; the operator is responsible for the content they generate.